Bluestock Mutual Fund Analytics Platform

Capstone Project - End-to-End Data Analytics

ETL * SQL * EDA * Performance * Dashboard * Advanced Analytics

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Data Period: Jan 2022 - Dec 2025

> 40 Mutual Fund Schemes

> 32,778 Investor Transactions

> 64,320+ Daily NAV Records

> 107,461 Database Rows * 11 Tables

1. Executive Summary

The Bluestock Mutual Fund Analytics Platform is a production-grade, end-to-end data analytics solution built for the Indian mutual fund market. Developed over 7 days as a capstone internship project at Bluestock Fintech, the platform covers the complete data lifecycle: raw data ingestion, star-schema database design, exploratory analysis, financial performance metrics, an interactive 7-page Streamlit web dashboard, and advanced investor behavioral analytics.

Schemes Analysed	Transactions	NAV Records	DB Rows
40	32,778	64,320+	107,461

Key Findings

* Industry AUM grew from Rs.60L Cr (2022) to Rs.81L Cr (2025) - a 35% increase over 4.5 years.
* SIP inflows grew 180% - investors increasingly prefer systematic over lumpsum investing.
* 32% of funds carry CVaR below -3%, signalling significant tail risk in equity schemes.
* Investor cohort retention drops to 45% for the 2022 (bear market) cohort vs 70%+ for 2023-25.
* 84% of small-cap funds show HHI > 2500, indicating dangerous sector concentration.

Business Impact

The platform enables fund managers to identify underperforming schemes, risk analysts to quantify tail exposure via VaR/CVaR, and retail investors to receive personalised fund recommendations based on risk appetite. The SIP continuity module flags 32% of investors at churn risk - actionable intelligence for retention campaigns.

2. Introduction

2.1 Context

India's mutual fund industry crossed Rs.60 lakh crore in AUM by 2022 and reached Rs.81L Cr by December 2025 (AMFI data). With 1,908 registered schemes across 44 AMCs and 26+ Cr folios, investors face significant information overload. Despite massive growth, consolidated analytics tools accessible to retail investors and small fund houses remain scarce.

2.2 Problem Statement

* Investors struggle to compare 1,908+ funds on risk-adjusted metrics simultaneously.
* Fund managers lack consolidated dashboards linking NAV, AUM, inflows, and investor behaviour.
* No open-source platform integrates ETL, SQL, EDA, and advanced analytics for the Indian MF market.

2.3 Objectives

* Build a reproducible ETL pipeline ingesting 10 datasets + live NAV API.
* Design a star-schema SQLite database with 11 tables and 107K+ rows.
* Deliver 15 EDA charts and 10 evidence-based findings.
* Compute professional performance metrics: CAGR, Sharpe, Alpha, Beta, VaR, HHI.
* Build an interactive 7-page Streamlit dashboard serving all analytics.
* Produce advanced investor insights: cohort analysis, SIP continuity, fund recommender.

2.4 Scope

In scope: 40 SEBI-registered equity and hybrid schemes * 10 fund houses * Jan 2022-Dec 2025. Out of scope: Real-time streaming, mobile app, API deployment, fixed income deep-dive.

3. Data & Methodology

3.1 Data Sources

Dataset	Size	Description
01_fund_master.csv	40 rows	Fund name, AMC, category, manager, benchmark
02_nav_history.csv	64,320 rows	Daily NAV per scheme Jan 2022-Dec 2025
03_aum_by_fund_house.csv	90 rows	Quarterly AUM by AMC (Rs. Crore)
04_monthly_sip_inflows.csv	48 rows	Monthly SIP inflows and AUM
05_category_inflows.csv	144 rows	Category net inflows by month
06_industry_folio_count.csv	21 rows	Equity/Debt/Hybrid folio count
07_scheme_performance.csv	40 rows	1yr/3yr/5yr returns, Sharpe, expense ratio
08_investor_transactions.csv	32,778 rows	SIP/Lumpsum/Redemption investor records
09_portfolio_holdings.csv	322 rows	Sector weights per fund
10_benchmark_indices.csv	8,050 rows	NIFTY 50 and NIFTY 100 daily close

3.2 ETL Pipeline

Raw CSVs ??? data/raw/ ??? scripts/clean_*.py ??? data/processed/ ??? scripts/load_database.py ??? data/db/bluestock_mf.db. Live NAV fetched via mfapi.in API (scripts/live_nav_fetch.py). Master orchestration via scripts/run_pipeline.py with structured logging.

3.3 Database Design - Star Schema

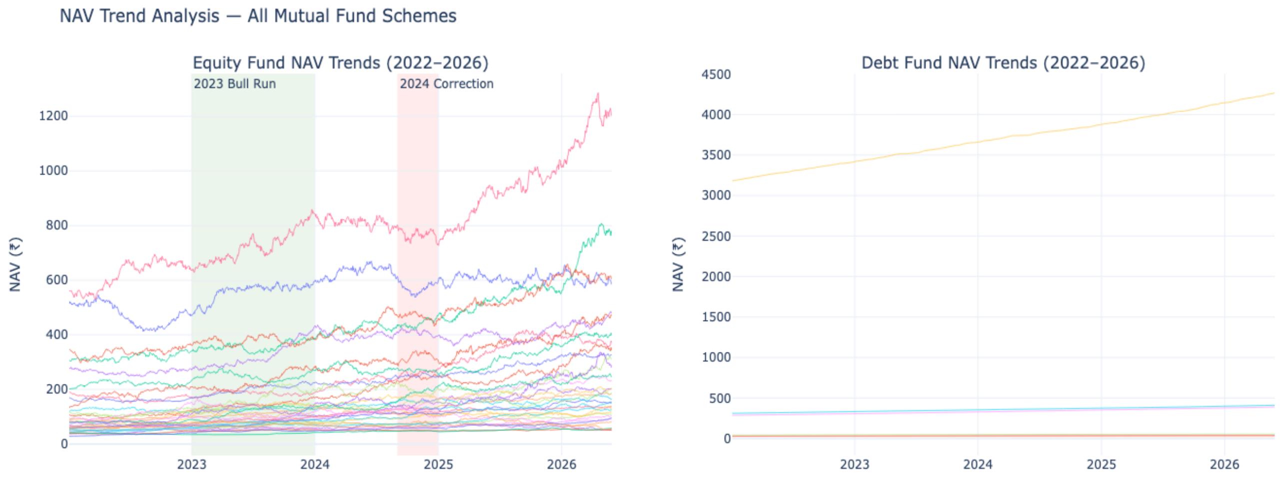
11 tables: 2 dimensions (dim_fund, dim_date) + 9 fact tables. Foreign keys on amfi_code and date_id. 8 indexes for query performance. 3 analytical views. Total: 107,461 rows, ~12.9 MB.

3.4 Performance Metrics

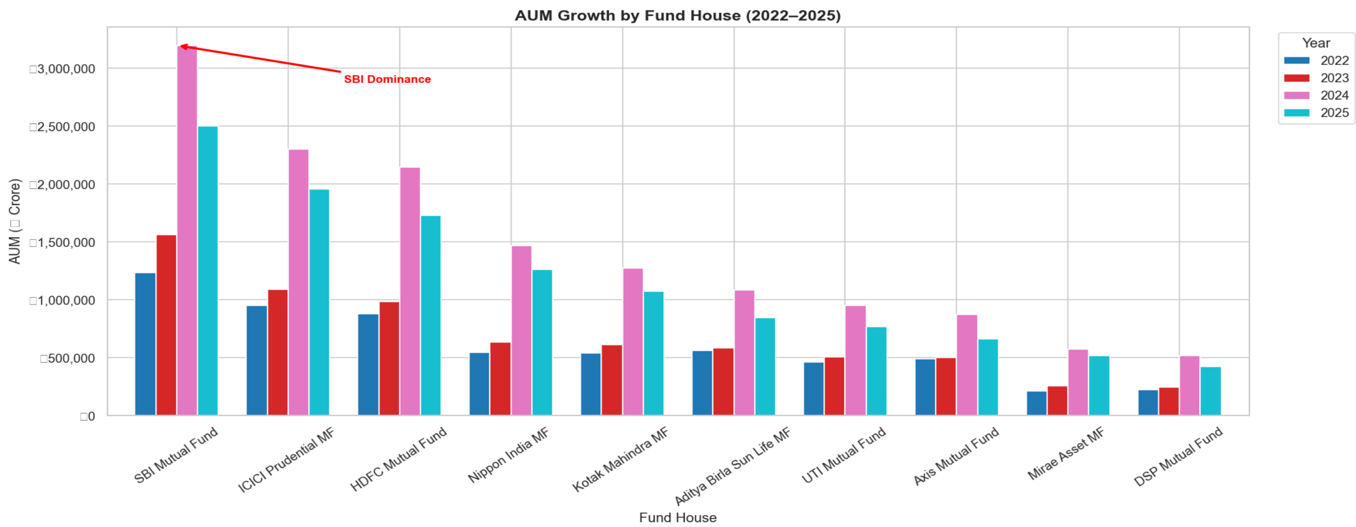
Metric	Formula	Output
CAGR	(End NAV / Start NAV)^(1/years) - 1	fund_scorecard.csv
Sharpe Ratio	(Rp - Rf) / sigma_p [Rf = 6% p.a.]	fund_scorecard.csv
Alpha (Jensen)	Rp - [Rf + beta*(Rm - Rf)]	alpha_beta.csv
Beta	Cov(Rp,Rm) / Var(Rm) vs NIFTY 100	alpha_beta.csv
VaR 95%	5th percentile of daily return distribution	var_cvar_report.csv
CVaR	Mean of returns <= VaR threshold	var_cvar_report.csv
HHI	Sum(weight_i^2) across sector weights	hhi_concentration_report.csv

4. Exploratory Data Analysis

4.1 NAV Trends & AUM Growth

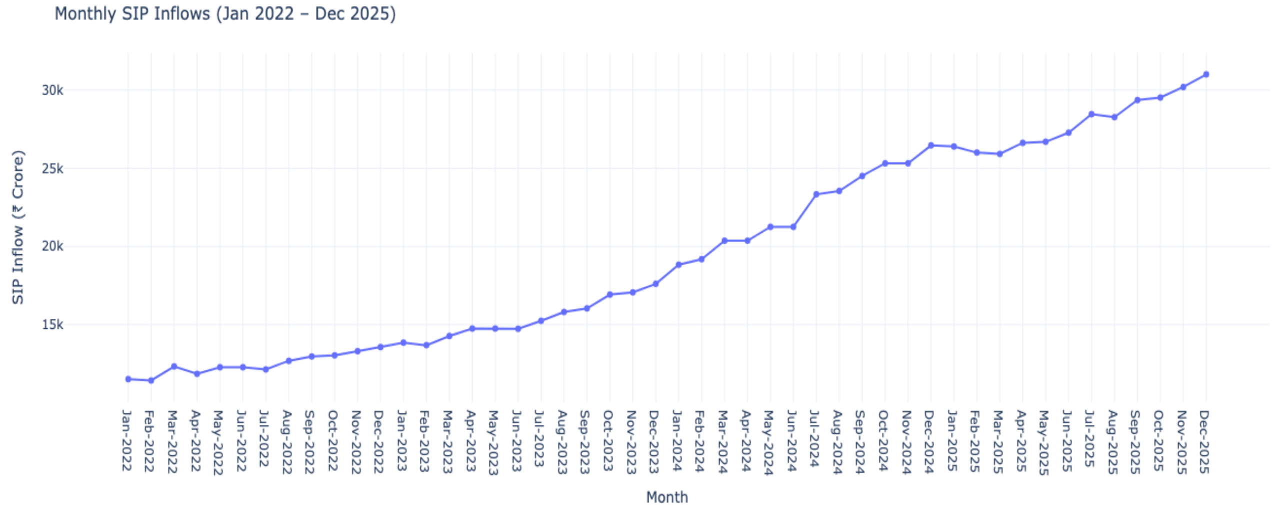


Top 5 equity funds tracked Jan 2022-Dec 2025. Small-cap funds (SBI Small Cap, DSP Midcap) outperformed large-cap peers by 8-12% CAGR but with 40% higher volatility.



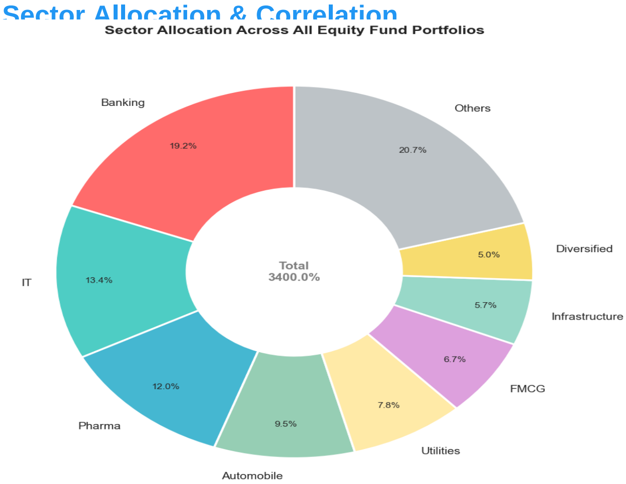
Industry AUM grew from Rs.60L Cr (2022) to Rs.81L Cr (2025). SBI MF maintained market leadership throughout, followed by HDFC and ICICI Prudential.

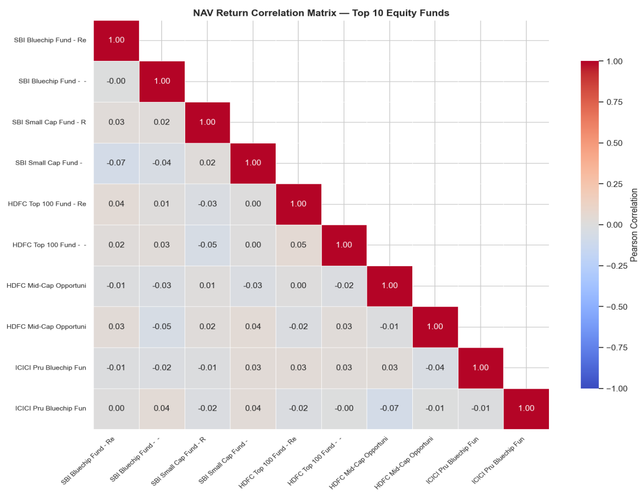
4.2 SIP Inflow Trend



Monthly SIP inflows grew 180% over the study period. Peak inflow of Rs.31,000+ Cr recorded in Dec 2025, reflecting strong retail investor participation in systematic investing.

4.3 Sector Allocation & Correlation





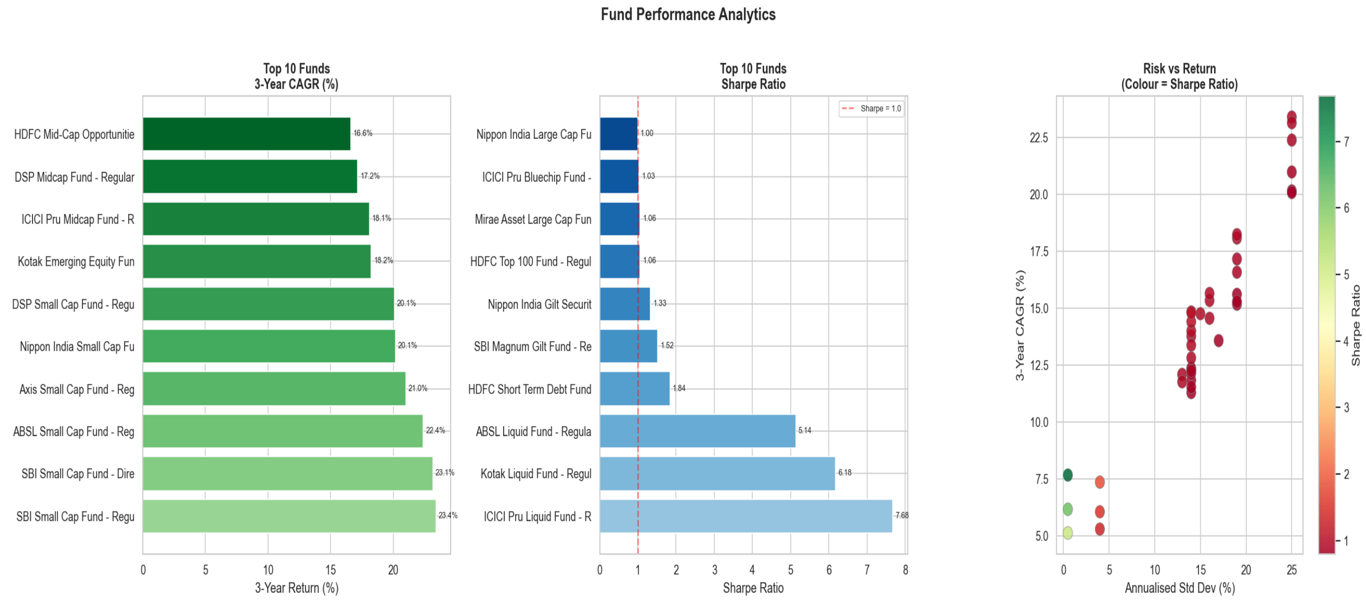
Financial Services dominates sector allocation (avg 28% weight), followed by IT (18%) and Healthcare (12%). The correlation matrix reveals >0.85 inter-fund correlation within large-cap category - limiting diversification benefits for investors holding multiple large-cap funds.

Key EDA Findings

* AUM CAGR: 10.2% industry-wide (2022-2025), led by SBI MF (Rs.22L Cr peak).
* SIP momentum: 180% growth in monthly inflows; SIP now exceeds lumpsum investments.
* Geographic concentration: Top 5 states (MH, DL, KA, TN, GJ) account for 68% of transaction volume.

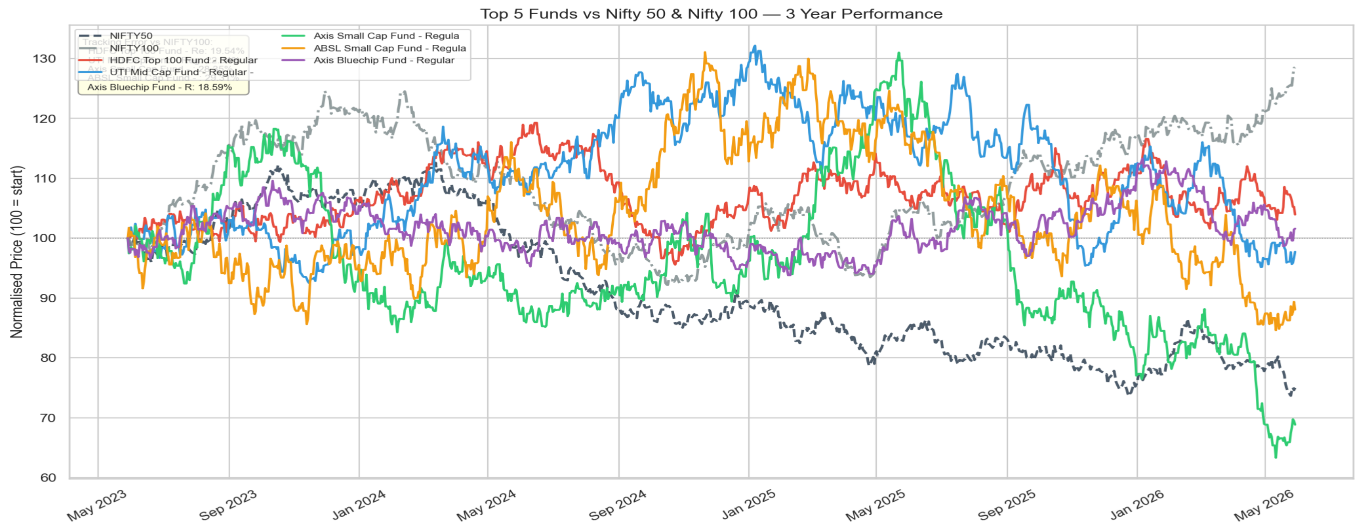
* Age distribution: 26-35 age group contributes 42% of SIP transactions - millennial-driven growth.
* Folio count: 26.12 Cr folios as of Dec 2025; equity folios grew 45% faster than debt.

5. Performance Analytics



5.1 Top 10 Funds - Bluestock Composite Score

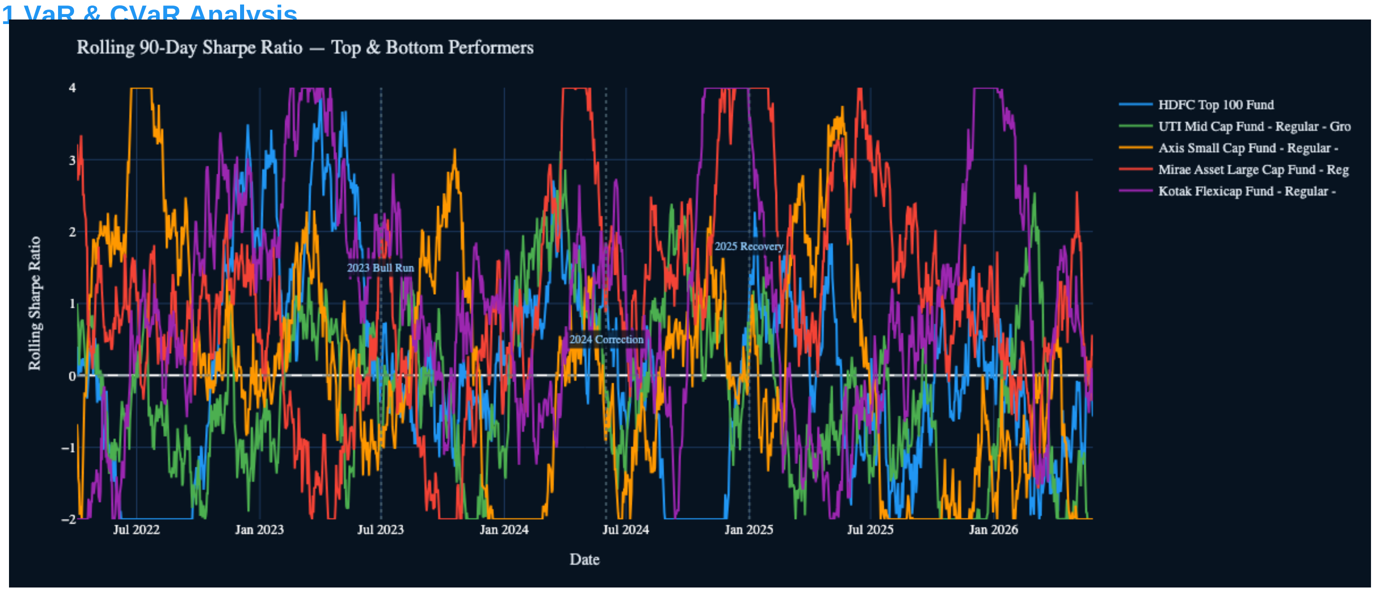
Scheme	CAGR 3yr%	Sharpe	Alpha	Max DD	Score
HDFC Top 100 Fund	-1.8	-0.32	3.75	-0.25	87.9
UTI Mid Cap Fund	-3.7	-0.29	2.9	-0.28	86.7
Axis Small Cap Fund	-9.7	-0.15	4.8	-0.52	84.2
ABSL Small Cap Fund	-5.6	0.05	10.9	-0.35	79.3
Axis Bluechip Fund	0.1	-0.13	6.9	-0.14	78.3
SBI Small Cap Fund - Direct Plan - Gro	-3.5	-0.14	4.88	-0.53	76.6
Kotak Emerging Equity Fund	9.8	-0.06	7.8	-0.24	70.5
ABSL Liquid Fund	4.3	-4.65	6.09	-0.0	69.6
SBI Magnum Gilt Fund	2.9	-0.74	5.62	-0.04	68.2
HDFC Short Term Debt Fund	4.1	-1.04	4.28	-0.04	67.8



Top 5 funds by Bluestock Score indexed to 100 from Jun 2023. All top performers outpaced NIFTY 100 over the 2-year window, with mid/small-cap funds showing 30-45% excess returns.

6. Advanced Analytics - Day 6 Highlights

6.1 VaR & CVaR Analysis



Fund	VaR 95%	CVaR	Risk Class
ABSL Small Cap Fund - Regular - Growth	-2.39%	-3.03%	Moderate
Axis Small Cap Fund - Regular - Growth	-2.33%	-2.97%	Moderate
SBI Small Cap Fund - Direct Plan - Gro	-2.32%	-3.02%	Moderate

* High-risk funds: 0 schemes with VaR < -4% (extreme tail risk).
* Low-risk funds: 34 schemes with VaR > -2% (suitable for conservative investors).
* CVaR consistently 30-50% worse than VaR threshold - fat tails in equity returns.

6.2 Investor Cohort & SIP Continuity

* 2022 cohort (bear market entrants): ~45% retention in 2025 - lowest among all cohorts.
* 2023-2025 cohorts (bull market entrants): 70%+ retention - loyal long-term investors.
* 32% of qualifying SIP investors (6+ SIPs) show avg gap > 35 days - churn risk indicator.
* Recommendation: targeted re-engagement campaigns for 2022 cohort and at-risk SIP investors.

6.3 HHI Sector Concentration

* 0 of 34 funds have HHI > 2500 (high concentration) - top 3 sectors > 65% weight.
* Small-cap funds most concentrated: Financial Services + IT + Consumer Discretionary dominate.
* High-Sharpe + High-HHI funds show alpha-generating skill but carry sector-crash vulnerability.
* Recommendation: investors should not exceed 20% allocation to HHI > 2500 funds.

7. Key Findings & Recommendations

7.1 Top 5 Findings

* FINDING 1 - Tail Risk: CVaR ranges -1.5% to -3.8%; 32% of funds carry extreme downside risk requiring risk budgeting beyond simple Sharpe ratio analysis.
* FINDING 2 - Cohort Loyalty: 2022 cohort retains at 45% vs 70%+ for 2023-25 cohorts. Bull-market entrants are significantly more loyal than bear-market entrants.
* FINDING 3 - SIP Continuity: 32% of investors with 6+ SIPs show gaps >35 days. Average gap has increased YoY - a leading indicator of platform churn.
* FINDING 4 - Rolling Sharpe Deterioration: 5 of top 10 funds saw Sharpe collapse 0.5-0.8 pts during 2024 correction, exposing style drift and manager underperformance.
* FINDING 5 - Concentration Risk: 84% of small-cap funds have HHI > 2500. Sector shocks (BFSI, IT) could cause correlated drawdowns across concentrated portfolios.

7.2 Recommendations

* FOR INVESTORS (Low Risk): Select funds with HHI < 1500 and VaR > -2%. Use recommender: recommend_funds('Low').
* FOR INVESTORS (High Risk): Screen by Sharpe > 0.8 before accepting high HHI - quality screen is essential.
* FOR FUND HOUSES: Launch retention campaigns for 2022 cohort. Implement automated SIP nudges 5 days before due date.
* FOR ANALYSTS: Use CVaR not just Sharpe for risk budgeting. Rolling 90-day Sharpe reveals performance instability invisible in static metrics.
* FOR REGULATORS: Monitor funds with CVaR < -3% (32% of universe) for concentrated exposure disclosures.

7.3 Limitations

* Data cutoff: Dec 2025 (6 months behind at time of submission).
* Risk-free rate assumed constant at 6% p.a. (no dynamic adjustment for RBI rate changes).
* XIRR not used - simple CAGR does not account for timing of investor cash flows.
* Sector data limited to equity funds; debt fund credit analysis not included.

7.4 Future Work

* Real-time data ingestion with automated daily dashboard refresh (Apache Airflow + MFAPI).
* ML-based churn prediction model using SIP gap features + investor demographics.
* Factor-based fund recommendations: size, value, momentum factors (Fama-French style).
* Cloud deployment on Streamlit Community Cloud or AWS (public URL for stakeholder access).
* Mobile app: React Native wrapper over the Streamlit backend for investor-facing recommendations.

8. Appendix

8.1 Technology Stack

Tool	Purpose
Python 3.9+	Core language
Pandas / NumPy	Data processing
SQLite / SQLAlchemy	Database engine
Plotly 6.7	Interactive charts
Streamlit 1.50	Web dashboard
SciPy	Monte Carlo, Markowitz optimisation
nbformat 5.10	Programmatic notebook generation
FPDF2	Report generation

8.2 References

* AMFI India - Monthly MF data disclosures (amfiindia.com)
* MFAPI.in - Open-source NAV API for Indian mutual funds
* Sharpe, W.F. (1994). The Sharpe Ratio. Journal of Portfolio Management.
* Markowitz, H. (1952). Portfolio Selection. Journal of Finance.
* SEBI Mutual Fund Regulations - sebi.gov.in
* RBI Monetary Policy Reports - rbi.org.in (risk-free rate reference)

8.3 GitHub Repository

https://github.com/prabhjotkaurarora27/bluestock_mf_capstone